L3.3 Linear dependence

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online B.Sc. program on data science.

Today's video is about linear dependence.

So, we have seen in the previous video what is the vector space, and in this video we

are going to study the notion of linear dependence of vectors.

So, first of all, let us recall what is vector addition.

So, vector addition is a binary operation plus from V cross V to V.

So, what that means is we take two vectors, v and w, and we can talk about the vector

v plus w.

So, in Rn we know that addition is defined coordinate wise.

So, x1, x2, xn plus y1, y2, yn is x1 plus y1, x2 plus y2 and so on up till xn plus yn.

So, just as an example, if you are 1.5 comma minus 3.3 comma 7.2 comma half comma 1 plus

minus 4 comma 5.8 comma 10 comma 5 halves comma minus 3.4, we have to add them coordinate

wise.

So, the first coordinate will be 1.5 minus 4 that gives you minus 2.5.

The second coordinate will be minus 3.3 plus 5.8 that gives me 2.5.

And I will leave you to check the rest.

We also have something called scalar multiplication in vector spaces, which means you, it is a

function from R cross V to V.

So, what that means is you take a scalar or a real number c and a vector v and then we

can associate to that a new vector c dot v. And so, remember this dot is corresponding

to scalar multiplication.

And it is standard to drop this dot, which is what we did in our previous video.

So, we are going to just use c times v, cv and not c dot v.

And in Rn, scalar multiplication is coordinate wise again.

So, c times x1, x2, xn is c times x1, c times x2, c times xn.

So, here is an example.

If you do 0.5 times the vector 1.5 comma minus 3.3 comma 7.2 comma half comma 1, that gives

me 0.75 comma minus 1.65 comma 3.6 comma one-fourth comma 0.55.

So, let us now look at what is the linear combination of vectors.

So, this is a key, I mean, this is a key definition which is going to drive this video and several

subsequent videos.

So, let us pay careful attention to this.

So, let V be a vector space and suppose we have a bunch of vectors v1, v2, vn.

So, the linear combination of v1, v2, vn with coefficients a1, a2, an, so these are real

numbers, is the vector summation aivi.

What does this mean?

This means you take vi, you take the ith vector vi, you do scalar multiplication of ai and

vi and that is a new vector.

So, accordingly, we have n new vectors, a1v1, a2v2, a3v3, anvn and then you add them up.

And remember, since our axioms tell us that, it does not matter which two we add first

what is called associativity of addition, you can just write it as a1v1 plus a2v2 plus

a3v3 up to anvn.

I do not need to specify which one I do first.

So, that is what we write as summation aivi.

So, this is a new vector.

And this new vector summation aivi is called the linear combination of v1, v2, vn with

coefficients a1, a2, an.

So, a vector v is a linear combination of v1, v2, vn if there are some scalars a1, a2,

an so that v is expressed as summation aivi.

So, let us do a couple of examples.

So, we are going to look at the geometry first and then we are going to look at the algebra.

So, the example here is 2 times 1 comma 2 plus 2 comma 1 is 4 comma 5.

So, what is 1 comma 2?

So, 1 comma 2 is over here.

What is 2 comma 1?

2 comma 1 is over here.

And then you take 2 times 1 comma 2, you scale 1 comma 2 by 2.

So, the new vector you get after scaling is this thing here which is 2 comma 4.

So, you have 2 comma 4 plus 2 comma 1, use the parallelogram law, and you get 4 comma

5.

So, of course, you can do this algebraically.

But that is not the point.

The point we want to make here is that we have expressed 4 comma 5 as a linear combination

of 1 comma 2 and 2 comma 1.

And what is the linear combination?

It is with the scalars 2 and 1.

So, 2 times 1 comma 2 plus 1 times 2 comma 1.

So, if there is a 1 time something, we suppress it.

So, 2 times 1 comma 2 plus 2 comma 1 is 4 comma 5 and the working out is clear.

And we can say more.

Each of the vectors in the expression is a linear combination of the other two vectors.

So, what are the three vectors we are talking about?

We are talking about 1 comma 2, 2 comma 1 and 4 comma 5.

We have expressed 4 comma 5 as a linear combination of 2 comma 1 and 1 comma 2.

So, instead, we could express 1 comma 2 as a linear combination of 4 comma 5 and 2 comma

1.

This is clear, because you can take 2 comma 1 on the other side, and you can divide by

half.

That is what we have got here.

So, the coefficients here are half and minus half.

And similarly, you can express 2 comma 1 as a linear combination of 4 comma 5 and 1 comma

2.

So, that is, to do that you take 1 comma 2 on the other side, and then you exactly get

the second expression written here.

So, we can do other things.

So, this is all some manipulation of expressions.

But this manipulation is important, because it is going to lead us to our definition.

So, you can rewrite this expression as 2 times 1 comma 2 plus 1 plus 2 comma 1, minus 4 comma

5 is 0 comma 0.

So, we have taken that first expression on the top and moved 4 comma 5 to the left by

subtracting it out.

And so what we get is on the right hand side, we get the vector 0.

So, we have a linear combination of these three vectors by some scalars and these are

non-zero scalars.

So, the coefficients here are nonzero.

What are the coefficients, 2, 1 and minus 1.

And with these coefficients, 0 is a linear combination of these three vectors, 1 comma

2, 2 comma 1 and 4 comma 5.

So, I will ask you to remember that that picture in the previous slide along with this observation.

Let us do one more.

So, here we have 3 times 2 comma 1 plus 2 times minus 2 comma 3 is 2 comma 9.

So, what is 2 comma 1, here is 2 comma 1.

And what is minus 2 comma 3, here is minus 2 comma 3.

So, if we scale these, so 3 times 2 comma 1 is exactly this large thing we get here,

which is 6 comma 3.

And 2 times minus 2 comma 3 is this larger vector we get here which is minus 4 comma

6.

And then when we add them, we use a parallelogram law and we get the vector 2 comma 9.

So, what have we done?

We have expressed 2 comma 9 as a linear combination of the vectors 2 comma 1, and minus 2 comma

3, the algebra is what we just did and the expression is written here.

So, now, again, we can take these three vectors 2 comma 1 minus 2 comma 3 and 2 comma 9 and

we can write any one of these as a linear combination of the other two.

So, already, you have written 2 comma 9 as a linear combination of the first two.

Now, here is how we write 2 comma 1 as a linear combination of minus 2 comma 3 and 2 comma

9.

Namely, we can take minus 2 comma 3 on the other side and then we can divide by 3, and

that is the expression we get here or we could write minus 2 comma 3 as a linear combination

of 2 comma 9 and 2 comma 1.

How do we do that?

We take 2 comma 1 on the other side and then we divide by half and indeed that is the expression

we get.

So, this is similar to what happened before.

And once again, instead of writing it this way, there is a third way we can write this

expression namely by taking 2 comma 9 on the other side and that gives us 0, the zero vector

on the right.

So, that is 3 times 2 comma 1 plus 2 times minus 2 comma 3 minus 2 comma 9 is 0 comma

0.

Again, what do we observe?

The 0 vector is a linear combination of 2 comma 1 minus 2 comma 3 and 2 comma 9 with

non-zero coefficients.

So, this is the important part.

What are the coefficients here?

The coefficients are 3, 2 and minus 1.

So, I should point out I am emphasizing these non-zero coefficients.

What happens if your coefficients are 0 if you take a linear combination of any vectors

v1, v2, vn with 0 efficient that is always going to give you 0, because the linear combination

will be 0 times v1 plus 0 times v2 plus 0 times vn which of course is the 0 vector.

So, the point is, here you can do it with non-zero coefficients.

So, you can always do it with 0 coefficients, but it is not clear you can do it with non-zero

coefficients.

And if you can, then there is something special happening.

So, remember, all these three vectors were on the same plane.

This is the key point.

Let us do an example in R3.

So, in R3 let us take these three, these two vectors.

So, we have 2 comma 2 comma 0.

So, this is a vector 2 comma 2 comma 0 right here.

So, this plane has been draw, I mean, this x, y, z axis has been drawn a bit differently

than we usually draw it for the purpose of illustration.

So, this is the XY plane.

So, on this we have the vector 2 comma 2 comma 0.

On the left side, this is the YZ plane, on this we have the vector 0 comma 2 comma 1.

And then we scale these.

So, if you scale this, you get the vector 0 comma 4 comma 2.

Here, we get the vector 3 comma 3 comma 0.

And you use a parallelogram and you add them and what you get is 3 comma 7 comma 2.

So, you can check that this is the, by projecting down to the exact plane what we get and by

projecting to the y-axis and so on that is what these other dashed lines are indicating.

So, what does this mean in terms of linear combinations?

So, in terms of linear combinations what we have seen is in R3 the vector 3, 7, 2 is a

linear combination of the vectors 0, 2, 1 and 2, 2, 0, and the coefficients were 2 and

3 halves.

So, that is the expression.

We just did that.

So, we can again play the same game.

So, instead of writing 3, 7, 2 as a linear combination of these two, we could instead

write, let us say, 0, 2, 1 as a linear combination of the other two, 3, 7, 2 and 2, 2, 0.

How do we do this?

Move the other term 1 to the right side and then divide by the corresponding coefficient

or you could write 2, 2, 0 as a linear combination of 3, 7, 2 and 0, 2, 1 again same idea.

So, I will suggest you check this.

And then again I can do the same thing that I have done before.

Namely, I can move the 3, 7, 2 in the original expression to the left hand side and get 0,

0, 0 on the right hand side.

So, what is the net result that the zero vector is a linear combination of 0, 2, 1, 2, 2,

0 and 3, 7, 2 with non-zero coefficients.

And again, I will encourage you to check that these three vectors lie on the same plane.

So, we can in fact get the equation of that plane.

So, the equation of the plane containing the two vectors 0, 2, 1 and 2, 2, 0 is 2x minus

2y plus 4z is 0.

And you can check that the third equation, the third point, indeed lies on this plane.

So, these three lie on the same plane.

And that is, so the geometry is somehow converting to this observation that if you take these

three vectors, you can write 0 as a linear combination in a non trivial way, meaning

with non-zero violations.

So, suppose we choose now a point or a vector which is not on the plane, say 1, 2, 0, so

1, 2, 0 is not on the plane.

How do I check 1, 2, 0 is not on the plane, I plug in these values into the equation of

the plane and get that it is not 0.

So, 2 times 1 minus 2 times 2 plus 4 times 0 is not equal to 0.

So, that is why it is not on the plane.

So, we claim then that 1, 2, 0 cannot be written as a linear combination of 0, 2, 1 and 2,

2, 0.

So, we saw earlier that, the earlier example that the third vector indeed was on the plane.

So, maybe we can go back and check what that was.

So, it was 3, 2, 7 and 3, 7, 2 does satisfy this equation, because 2 times 3 is 6, plus

4 times 2 is 8.

So, 6 plus 8 is 14, minus 2 times 7 is 0.

So, let us look at 1, 2, 0.

We saw already it is not, it does not satisfy that equation.

So, it is not on the plane.

And now the claim is it cannot be written as a linear combination of 0, 2, 1 and 2,

2, 0.

So, let us do this by contradiction.

Suppose you can write this as a linear combination, so you have a times 0, 2, 1 plus b times 2,

2, 0 is 1, 2, 0.

So, a and b are the coefficients.

Then we can work out what are the expressions in each coordinate.

So, the expressions in the first coordinate is just 2b, the expression in the second coordinate

is 2 times a plus 2 times b and the expression in the third coordinate is just a and we equate

it to the right hand side.

So, if we do that, then we get 2b is 1, 2a plus 2b is 2 and a is 0.

That means b is half and a is 0.

That is what we get from the first and third equations.

And if that happens, then 2a plus 2b cannot be 2.

So, that means there is no a and b which satisfies this which satisfies this equation.

So, our assumption that there is an a and b is wrong.

So, there is no such a and b.

So, what does that imply?

That implies that this vector 1, 2, 0 cannot be written as a linear combination of 0, 2,

1 and 2, 2, 0.

And we can equally well conclude from here and I will suggest that you check this that

if you do a times 0, 2, 1 plus b times 2, 2, 0 plus c times 1, 2, 0 is equal to 0, 0,

0.

Suppose I asked, can I get a, b and c such that it is 0, 0, 0, so what the above discussion

tells us is that the only possible choices for a, b, c are 0, 0 and 0, meaning all three

of a, b and c must be 0.

So, what does this mean?

So, this is now the other, well, the negation of the observation we did earlier.

So, the only way the zero vector is a linear combination of these three vectors is if the

coefficients are 0.

So, in the previous examples, where those vectors were on the same plane, we could write

0 as a linear combination of those three vectors with non-zero coefficients.

But here, there is no way of doing that.

The only way of writing 0 is if your coefficients are 0 and that is a trivial here, because

if the coefficient are 0 we certainly have, we get the 0 vector.

So, now we can define linear dependence.

Keep this geometry picture in mind and keep the algebraic expressions in mind.

A set of vectors v1, v2, vn from a vector space V is said to be linearly dependent,

if there exists scalars a1, a2, an not all 0 such that a1 times v1 plus a2 times v2 plus

an times vn is 0.

So, the key point here is that these a1, a2, an are not all 0.

So, there is a way of writing 0 as a linear combination where the coefficients are not

0.

At least some of them are not 0.

So, the equivalent way of saying this is that the zero vector is a linear combination of

v1, v2, vn with non-zero coefficients.

Meaning, when I say with non-zero coefficients that means some of the coefficients are not

0.

It is not, we do not need to say that all coefficients are non-zero, some non-zero coefficients.

Fine.

So, just to jog our memory, the first two, the first three examples we did, the two examples

in R2 and the third one in R3, the three vectors we had were linearly dependent.

That is what we saw, because we could write 0 as a linear combination of these three vectors

where the coefficients were non-zero, meaning not all of them were 0.

The fourth example meaning, where we had the vector 1, 2, 0 that example, the three vectors

in question, are not linearly dependent.

So, next video, we will see that they are what are called linearly independent.

So, consider the vectors 2, 3, 7 and five-third, five halves and 35 by 6 in R3.

So, you can check that if you multiply this second vector by 6 and you multiply the first

vector by 5, then you get the same you get the same vector.

So, in other words, if you do 5 times 2 comma 3 comma 7 minus 6 times five-third comma five

halves comma 35 by 6, that is 0, 0, 0.

So, at least one of these coefficients is non-zero.

In fact, in this case, both of them are non-zero.

And so these two vectors are linearly dependent.

These two vectors 2, 3, 7, and five-third, five halves, 35 by 6 are linearly dependent.

And note that the way this worked is that they are scalars, I mean, they are scalar

multiples of one another.

If you draw five-thirds, five halves, 35 by 6, and if you draw 2 comma 3 comma 7, they

will appear on the same line.

That is clear, because you scale one of them by 6 and the other by 5.

They give you the same vector.

So, in other words, when you have two vectors, what we are saying is if two vectors are linearly

dependent, then essentially one is a multiple of the other.

So, let us continue and study one more example.

So, consider the following three vectors in R3, 2, 1, 2, 3, 0, 1 and 10, minus 4 minus

2.

So, here is an equation that we can check easily, 2 times 2, 1, 2 minus 3 times 3, 0,

1 plus half times 10, minus 4, minus 2 is the 0 vector.

So, from here it follows that these three vectors are linearly dependent.

Let us do this checking just to be sure.

So, the checking here is 4 comma 2 comma 4, minus 9 comma 0 comma 3, plus 5 comma minus

2 comma minus 1.

Well, the first component is 4 minus 9 plus 5 that is indeed 0.

The second component is 2 minus 0, minus 2, which is 0.

And then the third component is 4 minus 3 minus 1, which is indeed 0.

So, this is indeed, this equation is indeed correct.

And this tells us that there is a linear equation satisfied by these three vectors with non-zero

coefficients which equates to 0.

So, they are linearly dependent.

Let us continue and append a vector to this set.

So, the set is now 2, 1, 2, 3, 0, 1, 10, minus 4 minus 2, and 2, 3, 7.

So, we have appended one more vector to this set.

And we can ask, is this set now linearly dependent?

Is this a set of linearly dependent vectors?

And the answer is, well, certainly it is.

Because since the original set, meaning the set with three of them was linearly dependent,

we can write down the same equation, but we also have to include the new vector 2, 3,

7.

So, what do we do, we put the 0 coefficient for that.

And we of course, still get 0, 0, 0.

So, you can, if you are not convinced, you can do this checking by yourself.

And from here it is clear that this set is linearly dependent.

So, what is the point?

The point is this we get the following fact.

I mean, this is by observation, but you can clearly see because the proof is the same.

If a set is linearly, if a set of vectors is linearly dependent, then so is every superset

of that set.

So, v1, v2, vn is linearly dependent, then v1, v2, vn and you add one more vector w,

then v1, v2, vn comma w is also linearly dependent.

You can add as many vectors as you want.

They are all linearly dependent.

So, I guess that finishes what, the material in this video.

Let us quickly recall that we have seen in this video that when a set of vectors in a

vector space is linearly dependent, and importantly, the, in the case of R2 or R3 what this means

is that if you have two vectors are linearly dependent, they must be on the same line.

If you have three vectors are linearly dependent, they must be on the same plane.

This is the geometry of what is going on.

And finally, to check these things, we have to solve some equations.

So, finally, this kind of checking is also related to solving systems of linear equations.

So, thank you.